MCQ solutions

1. **A.** By linearity, $E(Y) = 2E(X) - 1 = 5$. The constant is subtracted once.
2. **C.** The variance is multiplied by $(-3)^2 = 9$, and a translation does not change it. It cannot be negative.
3. **B.** The sum has variance $n\sigma^2$, then division by n multiplies the variance by $1/n^2$.
4. **D.** Linearity of expectation does not require independence. Additivity of variances requires additional conditions.
5. **C.** $E(X) = 1$, $E(X^2) = 2$, so $V(X) = 2 - 1^2 = 1$.
6. **A.** It is an upper bound on a deviation from the expectation, not an equality nor a bound applying directly on X.
7. **D.** Solve $25/n \leq 0.05$ with $n > 0$ gives $n \geq 500$. This is a sufficient size according to this bound.
8. **B.** A probability is always at most 1; the upper bound 2 gives no improvement.
9. **C.** This is convergence in probability, not a decrease of the error at each observa-
10. **A.** The mean is exactly X. Repeating the same variable does not create new observations independent.

True/false. False: the upper bound may be crude. True: take the square root of $a^2V(X)$. False: question 10, where the variance does not decrease.

Solutions to the applications and guidance

Application 1. Sum of probabilities 1. E=2; $E(X^2) = 11/2$, therefore $V = 3/2$ and $\sigma = \sqrt{6}/2$.

Application 2. $E(Y) = 4$, $V(Y) = 27/2$; $E(Z) = -2$, $V(Z) = 3/2$. The squared deviations from the mean does not change under sign inversion.

Application 3. Independence: variance 1 $/(4n)$. Identical copies: the mean equals X, so the variance 1/4 whatever n. Repeating the same measurement does not create an independent sample.

Application 4. The bound is 9/ $(0{,}25n) = 36/n$. Requiring $36/n \leq 0.01$ gives $n \geq 3600$.

Application 5. The event is $X \leq 2$ or $X \geq 8$. By symmetry, its probability is 2 $(1 + 10 + 45)/2^{10} = 7/64 \simeq 0{,}109375$. The bound equals $(1/40)/(9/100) = 5/18 \simeq 0{,}2778$, so it is correct but less precise.

Application 6. Theoretical standard deviations are about 0 ,1581, 0,05 and 0,01581. The simulated outputs fluctuate: we are comparing a set of means, not the successive values of a single mean.
Empirical proportions are not proofs. The reproducible results file
gives the values obtained with the chosen seed.

Application 7. The results 0, 1 give frequency 1 /2 ; add 1 gives 2 /3, further from 1 /2. For every fixed accuracy, the probability of a deviation at least equal to this accuracy tends to 0 as n tends to infinity. This does not impose a monotonic approach on each realisation.

Reading the guarantees

Have students circle the strict inequality of the central event and the non-strict one of its complement. Compare systematically a computable probability and a universal bound. Do not present the simulation as a proof of the law of large numbers.

Teaching notes and solutions

Guidance for the teacher

Written test 5 and the second mock exam come before S73: ask for neither Bienaymé-Chebyshev nor the distribution of large numbers. For the variance, have students circle the independence assumption. For the concentration, distinguish exact probability from upper bound, then sufficient index from minimal index real. A simulation is not a proof, and an observed mean does not approach necessarily with each new trial.

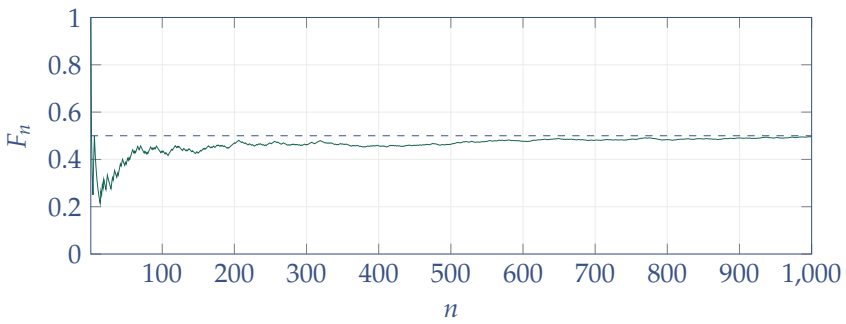
Solutions to the activity

1.

```
from random import Random

def trajectoire_piece(N=1000, graine=2026):
    rng = Random(graine)
    piles, frequences = 0, []
    for n in range (1, N+1):
        piles += rng.random() < 0.5
        frequences.append(piles/n)
    return frequences
```

Plot the points (n, F_n) with the list obtained. After two trials giving one head and one tail, $F_2 = \frac{1}{2}$. Whatever the third outcome, F_3 value $\frac{1}{3}$ or $\frac{2}{3}$: the gap to $\frac{1}{2}$ increases. The stabilisation is not a monotonic decrease of the deviation along each path.



A simulated path, seed 2026; the line is $y = \frac{1}{2}$.

2. Let us denote B_i the indicator of heads. Then $F_n = \frac{B_1+\cdots+B_n}{n}$. Each B_i has expectation $\frac{1}{2}$ and variance $\frac{1}{4}$. By linearity, $E(F_n) = \frac{1}{2}$; by independence and change of scale, $V(F_n) = \frac{n(\frac{1}{4})}{n^2} = \frac{1}{4n}$.

3. Bienaymé-Chebyshev gives $P(|F_{1000} - \frac{1}{2}| \geq 0{,}05) \leq \frac{1}{4\times 1000\times 0{,}05^2} = 0{,}1$. It is a bound upper, not of the exact probability.

4. To ensure a complementary probability of at most 0 ,01, it suffices to $\frac{1}{4n\times 0{,}05^2} \leq 0{,}01$. This is equivalent to to $n \geq 10000$. Then $P(|F_n - \frac{1}{2}| < 0{,}05) \geq 0{,}99$.

5. The simulation illustrates one finite realisation; other seeds give other trajectories. The in- establishes a guarantee on the probability, for any sample satisfying the hypotheses of independence and identical distribution.

E1. The affine transformation gives $E(4X-1) = 4E(X) - 1 = 11$ and $V(4X-1) = 4^2 V(X) = 32$. The standard deviation is $\sqrt{32} = 4\sqrt{2}$; the shift -1 does not change the spread.

E2. By linearity, $E(2X-Y) = 2E(X) - E(Y) = 4 - 5 = -1$. The independence of X, Y allows of adding the variances of 2 X and $-Y$: $V(2X-Y) = 4V(X) + V(Y) = 4 + 4 = 8$.

E3. The mean of 64 independent identically distributed variables keeps expectation 10. Its variance is $\frac{4^2}{64} = \frac{1}{4}$, so its standard deviation is $\frac{1}{2}$.

E4. Bienaymé-Chebyshev with $\delta = 6$ gives $P(|X-100| \geq 6) \leq \frac{V(X)}{6^2} = \frac{9}{36} = \frac{1}{4}$. This bound does not does not give an equality or an exact value.

E5. We have $P(|M_n - \mu| \geq 0{,}2) \leq \frac{4}{n \times 0{,}2^2} = \frac{100}{n}$. For this bound to be at most 0 ,05, it suffices to $n \geq 2000$. The complement then has probability at least 0.95.

Solutions to homework DM14

1. Let us denote B_i the independent success indicators. $F_n = (\frac{1}{n}) \sum_{i=1}^{n} B_i$, with $E(B_i) = p$, $V(B_i) = p(1-p)$. Hence

$$E(F_n) = \frac{1}{n} \sum_{i=1}^{n} p = p, \qquad V(F_n) = \frac{1}{n^2} \sum_{i=1}^{n} p(1-p) = \frac{p(1-p)}{n}.$$

2. $p(1-p) = p - p^2 = \frac{1}{4} - (p - \frac{1}{2})^2 \leq \frac{1}{4}$. Equality is attained for $p = \frac{1}{2}$.

3. For every $p \in [0,1]$, $P(|F_n - p| \geq 0{,}02) \leq \frac{1}{4n \times 0{,}02^2}$. It suffices that this bound is at most 0 ,05, i.e. $n \geq 12500$. By the complement, this size guarantees the required probability, uniformly in p.

4. On $[0, 0{,}1]$, the function $p \mapsto p(1-p)$ is increasing because its derivative 1 $- 2p$ is positive. So $p(1-p) \leq 0{,}1 \times 0{,}9 = 0{,}09$. It now suffices to $n \geq \frac{0{,}09}{0{,}05 \times 0{,}02^2} = 4500$. These sizes are sufficient in the sense of the upper bound, not necessarily minimal for the exact probability.

5. If all trials copy the same variable B, then $F_n = \frac{B + \cdots + B}{n} = B$. Hence $V(F_n) = p(1-p)$, without division by n. Adding variances was allowed only because of independence; this hypothesis disappears here. For $0 < p < 1$, increasing the number of copies brings no new information.

Solutions to the textbook or past-paper work

M_N is the average number of universal donors per sample of 100 people. Linearity gives $E(M_N) = 7.14$; independence gives $V(M_N) = \frac{6{,}63}{N}$.

$$P(|M_N - 7{,}14| \geq 0{,}14) \leq \frac{6{,}63}{0{,}14^2 N}.$$

To make this bound at most 0.05, it is enough to take

$$N \geq \left\lceil \frac{6{,}63}{0{,}05 \times 0{,}14^2} \right\rceil = 6766.$$

By complement, $P(7 < M_N < 7{,}28) \geq 0{,}95$.